WORKED SOLUTION

STEP II 2004, Question 1

Pure • Equations with Surds

STEP II 2004, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s2-q1

The key idea

Every equation involves $\sqrt{x}$ and $\sqrt{3x^2+1}$, so put $t=\sqrt{x}$ (with $t \geq 0$) to turn the first into a polynomial. Then isolate the remaining square root and square once. The result is a quartic in t, and it factorises easily because $t=0$ and $t=1$ are always roots. The real work is at the end: squaring can introduce false roots, so we keep only those for which the side we squared was not negative.

Solution

All three equations need $x \geq 0$ for $\sqrt{x}$ to be real, so we write $x=t^2$ with $t \geq 0$. Then $\sqrt{x}=t$ and $3x^2+1=3t^4+1$.

The squaring step works in only one direction. For real A and B,

$$\sqrt{A}=B \iff A=B^2 \text{ and } B \geq 0.$$

So after squaring we must check that the right-hand side is non-negative at each root.

Part (i)

Isolating the root gives

$$\sqrt{3t^4+1}=2t^2-t+1.$$

The right-hand side is always positive: its discriminant is $1-8<0$ and its leading coefficient is positive. So squaring loses nothing here, and every root of the squared equation is a genuine solution. Squaring,

$$\begin{aligned} 3t^4+1 &= (2t^2-t+1)^2 = 4t^4-4t^3+5t^2-2t+1, \\ 0 &= t^4-4t^3+5t^2-2t = t\,(t^3-4t^2+5t-2). \end{aligned}$$

The cubic vanishes at $t=1$ (since $1-4+5-2=0$), and dividing out the factor $t-1$ gives

$$t^3-4t^2+5t-2=(t-1)(t^2-3t+2)=(t-1)^2(t-2).$$

So $t\,(t-1)^2(t-2)=0$, giving $t=0,1,2$. All three are non-negative, so

$$x=0,\quad x=1,\quad x=4.$$

As a check, at $x=4$ we get $\sqrt{49}+2-8-1=0$.

Part (ii)

Now we have

$$\sqrt{3t^4+1}=-t^2+2t+1.$$

Squaring,

$$3t^4+1=(-t^2+2t+1)^2=t^4-4t^3+2t^2+4t+1,$$
$$0=2t^4+4t^3-2t^2-4t=2t\,(t^3+2t^2-t-2).$$

Grouping the terms of the cubic as $t^2(t+2)-(t+2)$ gives

$$2t\,(t+2)(t-1)(t+1)=0.$$

Since $t\geq 0$, only $t=0$ and $t=1$ remain. The right-hand side $-t^2+2t+1$ takes the values 1 and 2 there, both positive, so both roots are genuine:

$$x=0,\quad x=1.$$

Part (iii)

This time

$$\sqrt{3t^4+1}=t^2+2t-1.$$

Squaring,

$$3t^4+1=(t^2+2t-1)^2=t^4+4t^3+2t^2-4t+1,$$
$$0=2t^4-4t^3-2t^2+4t=2t\,(t^3-2t^2-t+2).$$

Grouping the cubic as $t^2(t-2)-(t-2)$ gives

$$2t\,(t-2)(t-1)(t+1)=0.$$

The non-negative roots are $t=0,1,2$.

Ask yourself Do all three of these values satisfy the original equation?

No. At $t=0$ the right-hand side t^2+2t-1 equals -1, but a square root can't be negative. So $t=0$ is a false root, introduced by the squaring. (Directly: at $x=0$ the left-hand side of (iii) is $1-0-0+1=2\neq 0$.) At $t=1$ and $t=2$ the right-hand side is 2 and 7, both positive, so these roots are genuine. Hence

$$x=1,\quad x=4.$$

There is a neat link between (ii) and (iii). Replacing t by $-t$ turns $-t^2+2t+1$ into $-(t^2+2t-1)$, and squaring removes the minus sign. So the squared equation in (iii) is the squared equation in (ii) with t replaced by $-t$, and its roots $0,\pm 1,2$ are the negatives of the roots $0,\pm 1,-2$ from (ii).

Checks

Substitute each answer into the original equation. For example, $x = 4$ in (iii) gives $\sqrt{49} - 4 - 4 + 1 = 0$. A fine numerical scan of each left-hand side for $0 \leq x \leq 50$ found no other zeros. In (i) the left-hand side touches zero at $x = 1$ without changing sign, which matches the double factor $(t-1)^2$.

Common slips

- Forgetting to check the sign of the side being squared, and so keeping $x = 0$ in (iii).
- Dividing by t (or by $\sqrt{x}$) and so losing the root $x = 0$ in (i) and (ii).
- Squaring with two square roots still on one side. Isolate $\sqrt{3x^2+1}$ first, so that a single squaring removes it.

Beyond STEP

The map $A \mapsto A^2$ is not injective on $\mathbb{R}$, so squaring both sides gives an implication, not an equivalence. Keeping track of which steps are reversible is the habit behind every careful solution of an equation, and in analysis it becomes the question of when a function has an inverse and on what domain. Bartle and Sherbert, *Introduction to Real Analysis*, sets this out carefully.

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